

GATORQUANTHACKS · DEMO

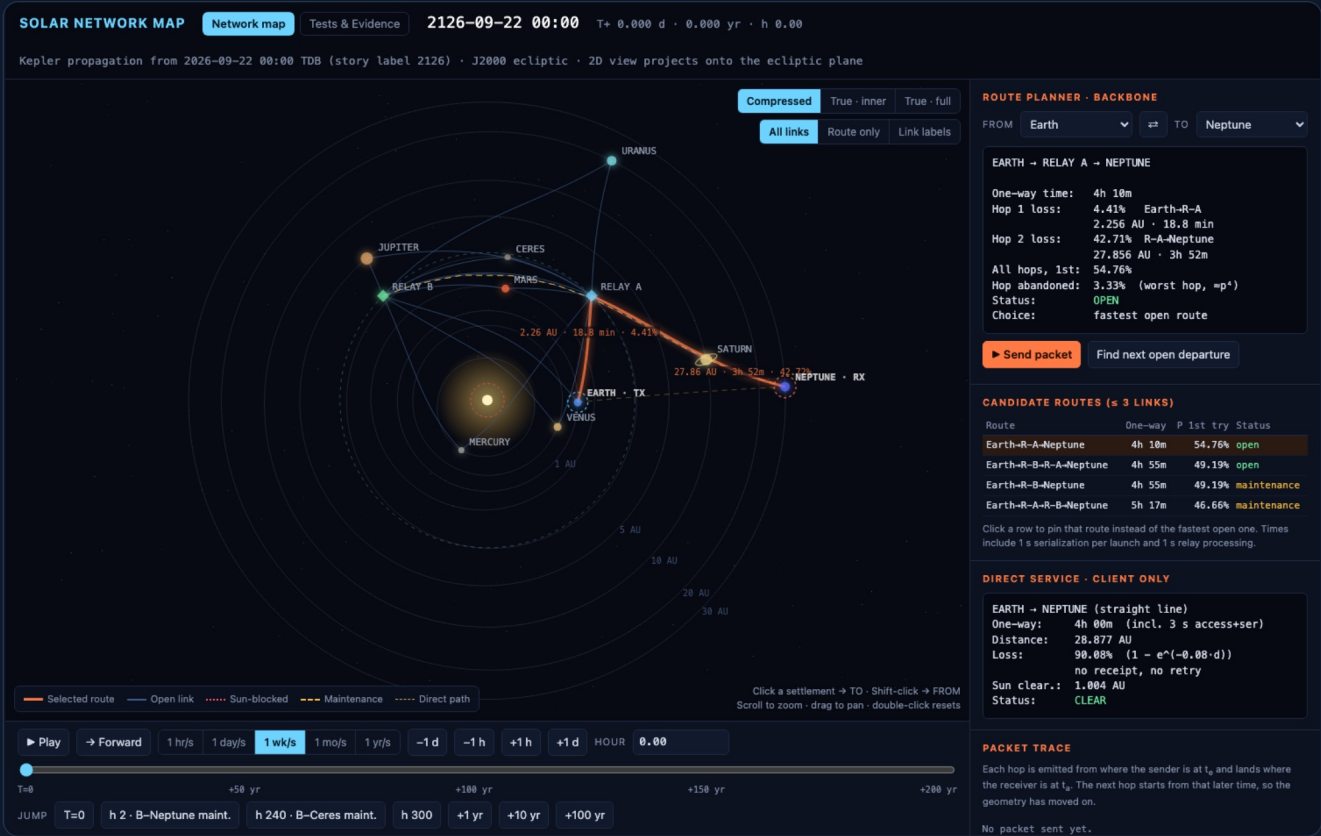
MultiPlanetary Exchange

What does an exchange look like when information takes hours to move?

Orbital network model → market engine → evidence framework. Clearing house on Ceres.



Delays are computed, not assumed



Planet motion → candidate routes → receiver position at arrival → relay choice → Sun exclusion → maintenance → delay & loss

4 h 10 m

Earth → Relay A → Neptune at hour 0

54.76%

arrive first try; the 27.9 AU hop loses 42.7% of launches

+1 yr

new geometry, new routes, new numbers

WHY CERES HOSTS THE CLEARING HOUSE

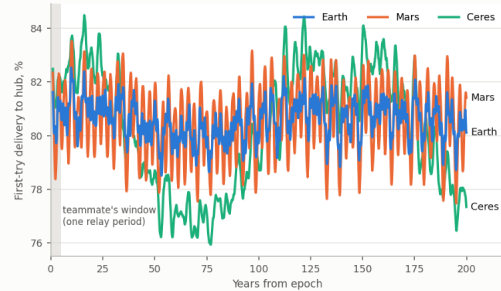
Latency ties. Reliability decides.

Clearing house over 200 years: delivery probability and Sun blockage together

Ceres delivers best for its first ~30 years, then trails for ~70. Earth is steady, never passes the 30-day packet lifetime, and ties Mars on delivery.

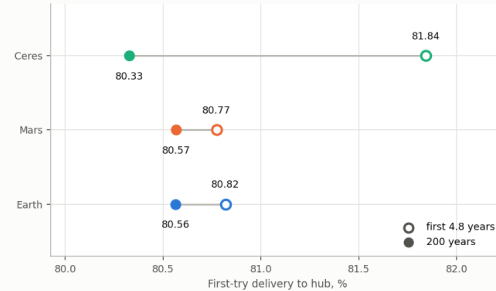
A Delivery probability by year

Mean over the 8 other settlements of the best open route's first-try success (1-year rolling)



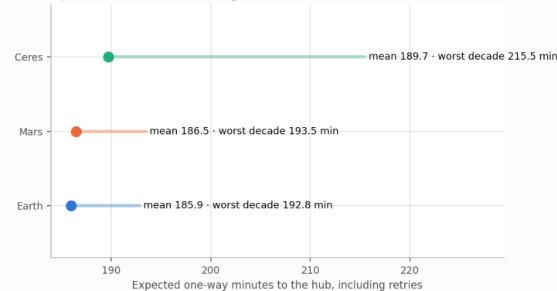
B The window decides the ranking

Ceres leads in one relay period; over 200 years Earth and Mars lead (differences < 0.3 points)



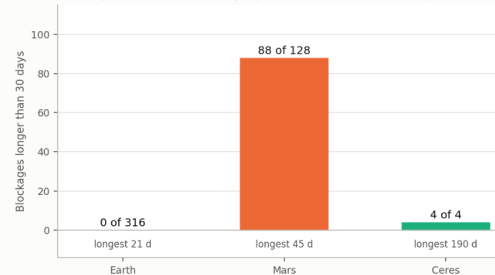
C Delivery loss turned into delay

Light time + expected retry waits (R_h per lost launch); lower is better



D Blockages that outlast the packet lifetime

Packets queued on the blocked relay expire; sessions must be re-routed (quota cost)



Source: network/python/hub_tradeoff.py on Data/orbital_elements.json (30-day samples for delivery and time; daily Sun test for blockages).

LATENCY

38.6 h vs 38.5 h

Cross-planet trade, Earth hub vs Ceres hub

RELIABILITY, 200 YEARS

8.6% → 0.8%

time down to one relay · re-routes 316 → 4

ACCEPTED COST

+31% backbone packets with today's Earth-heavy accounts;
rare long Ceres blockages

Reserve first, then cross the network

- 1 · Home exchange

Local trades clear on the local ledger
- 2 · Reserve first

Cash or shares locked before the order leaves
- 3 · Ceres batch

Cross-planet orders clear by fixed auction rules
- 4 · DvP at home

Both legs settle on the home ledgers

EARTH → NEPTUNE SHARE TRADE

E-Alice · 100 shares
to N-Eve at Neptune

↓

RESERVED h 1.0003
BOR-000001 locks at Earth

↓

BATCH EXECUTED h 31.1844
100 @ 47.50

↓

LOCKED → IN TRANSIT h 31.7385
XFR-000001 leaves Earth

↓

PACKET SENT h 31.7387
Earth → Relay A

↓

RELAY A h 32.0492
relay stores, receipts, forwards first copy

↓

NEPTUNE RECEIVED h 35.9124
data delivered

↓

VALIDATED · FINAL h 35.9124
usable by N-Eve at Neptune

Route	Earth → Relay A → Neptune
Distance	30.1052 AU (first successful copy per hop)
Send time	h 31.7387 T+31:44:19
Arrival time	h 35.9124 T+35:54:45
Total delay	34.91 h (order stamp → usable)
Retries	0
Packet losses	0
Transaction ID	XFR-000001 (leg of TRD-000001)
Duplicates ignored	0
E-Alice cash	\$70,000.00 → \$74,750.00
E-Alice shares	600 → 500
N-Eve cash	\$55,000.00 → \$50,250.00
N-Eve shares	500 → 600
Final state	COMPLETED · other leg XFR-000002 (cash) final h 39.4642

- h 1.0 · reserved**

Alice 100 sh ≥ \$45 · Eve ≤ \$50
- h 31.18 · \$47.50**

Midpoint; Eve's spare \$250 freed
- h 35.91 · shares**

Final and usable at Neptune
- h 39.46 · cash**

\$4,750 usable at Earth

Being closer to Ceres doesn't buy priority

GLOBAL BATCH AUCTION

Same price 100: arrived at the market E-Bob → N-Eve; priority by source time N-Eve → E-Bob. Network arrival time did not decide.

BAT-000001 · opens h 9.5000 · closes h 40.1870 · clears @ 100.00

M-Carla @ Mars · SELL 10 @ min 100
source h 10.0003 · arrived h 10.4959

E-Bob @ Earth · BUY 10 @ max 100
source h 10.0169 · arrived h 10.5725

N-Eve @ Neptune · BUY 10 @ max 100
source h 10.0003 · arrived h 14.1057

GLOBAL BATCH
@ Ceres
↓ match
@ 100.00

✓ M-Carla: 10 / 10 · rank 1
priority rank 1: filled in full

✗ E-Bob: 0 / 10 · rank 2
priority rank 2: outranked, quantity taken by better price or earlier source time

✓ N-Eve: 10 / 10 · rank 1
priority rank 1: filled in full

Batch length = 1.1 × slowest one-way (Neptune 4.11 h) + grace for 3 hop retries (26.17 h) = 30.69 h.

Order	Account	Side	Qty	Limit	Source time	Arrival	Allocated	Reason
BOR-000002	M-Carla @ Mars	sell	10	100	h 10.0003	h 10.4959	10	priority rank 1: filled in full
BOR-000003	E-Bob @ Earth	buy	10	100	h 10.0169	h 10.5725	0	priority rank 2: outranked, quantity taken by better price or earlier source time
BOR-000001	N-Eve @ Neptune	buy	10	100	h 10.0003	h 14.1057	10	priority rank 1: filled in full

► Clearing table (price → demand / supply / volume)

Bob (Earth) and Eve (Neptune) both bid \$100 for the last 10 shares.

Bob's order arrives **3.5 h earlier**. Eve sent hers **1 minute earlier**.

Eve 10 · Bob 0

Counterfactual, run once by hand: with arrival-time priority this test fails.

Rule: best price → earlier source stamp from the home exchange → exact ties pro-rata (spare unit by public hash) · one midpoint clearing price

05 · 3:35–4:15 · Present from slide (optional live: Equal-price auction → Global batch auction)

ATTACK 1 · LOSE A PACKET, THEN REPEAT IT

Exactly one financial effect

EARTH → NEPTUNE SHARE TRADE	
E-Alice · 100 shares to N-Eve at Neptune	
↓	
RESERVED h 1.0003 BOR-000001 locks at Earth	
↓	
BATCH EXECUTED h 31.1844 100 @ 47.50	
↓	
LOCKED → IN TRANSIT h 31.7385 XFR-000001 leaves Earth	
↓	
PACKET SENT h 31.7387 Earth → Relay A	
↓	
LOST (Launch #1) h 32.0498 Relay A → Neptune: forced loss (test): first launch of the share transfer on Relay A -> Neptune	
↓	
RETRY #2 h 40.7753 Relay A → Neptune	
↓	
RELAY A h 32.0492 relay stores, receipts, forwards first copy	
↓	
NEPTUNE RECEIVED h 44.6382 data delivered	
↓	
VALIDATED · FINAL h 44.6382 usable by N-Eve at Neptune	
↓	
SOURCE RECONCILED h 48.8116 Earth confirms	

Route	Earth → Relay A → Neptune
Distance	30.1078 AU (first successful copy per hop)
Send time	h 31.7387 T+31:44:19
Arrival time	h 44.6382 T+44:38:18
Total delay	43.64 h (order stamp → usable)
Retries	1
Packet losses	1
Transaction ID	XFR-000001 (leg of TRD-000001)
Duplicates ignored	2
E-Alice cash	\$70,000.00 → \$74,750.00
E-Alice shares	600 → 500
N-Eve cash	\$55,000.00 → \$50,250.00
N-Eve shares	500 → 600
Final state	COMPLETED

Lost → retried

Launch 1 forced lost at Relay A; retry 2 carries the same transaction ID

2 duplicates ignored

They return the existing record; Eve credited 600 shares once

4 of 4 launches

Worst-case order still lands before the batch closes

Still checked after every step: no double spend · no duplicate settlement · cash and shares conserved

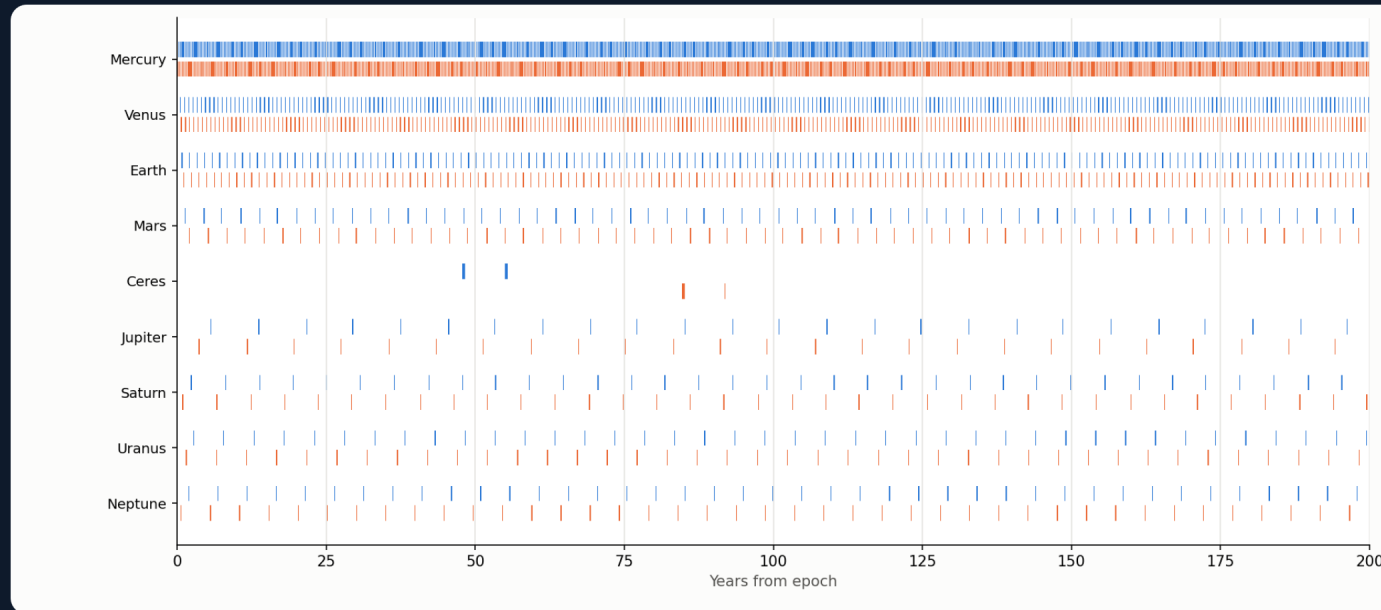
ATTACK 2 · A FUTURES TRADER STOPS PAYING

The winner is paid in full



LONG-HORIZON EVIDENCE · WAS THIS CHERRY-PICKED?

Not cherry-picked: 200 years of orbits



5,748 boundaries

Sun-blockage starts and ends found in the 200-year scan, each refined to 1 s; all flip exactly there

0 days

any settlement loses both relays

+1 / +10 / +100 y

batches 34.1 / 31.4 / 33.3 h, same rules

72 h Ceres outage

local trading continues; service recovers

Sun closures by link, 200 years · blue: Relay A blocked · orange: Relay B blocked

TESTS PASSED ✓ 17 / 17 scenarios on the real engine + network	FINANCIAL INVARIANTS ✓ PASS 10,146 checks, one after every step	CASH CONSERVED ✓ PASS ledgers + in transit = opening	SHARES CONSERVED ✓ PASS ledgers + in transit = opening	DOUBLE SPENDING ✓ NONE 6 attempts tried and blocked	DUPLICATE SETTLEMENTS ✓ NONE 6 repeat deliveries safely ignored
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17/17 test scenarios pass. Evidence adds 1 rising-price futures run: 18/18 evidence runs conserve cash and shares at every step.

WHAT WE PROVED

A fair, solvent market across the solar system

17 / 17

test scenarios pass on
the real orbits

0

double spends or
duplicate settlements

\$0

unbacked in the
tested default

0.8%

single-relay exposure
at Ceres (Earth 8.6%)

18 / 18 evidence runs (the 17 tests + 1 rising-price futures run) conserve cash and shares at every step.

Implemented: retries, deduplication, ordinary recovery. **Designed, not yet built:** proactive relay re-pinning and resubmission; at one difficult epoch a funded future stays open past maturity (E5).

Physics → market → settlement → evidence. All of it runs.